

Overview of the ScBridge workflow

Input and Data Processing

FASTQ Files

```
@SEQ_ID  
ACTF...  
+  
IIIIII...
```

Expression Matrix

Gene	Cell1	Cell2	Cell3
Gene1	12	0	8
Gene2	0	6	5
Gene3	3	0	15
...			

Seurat Object

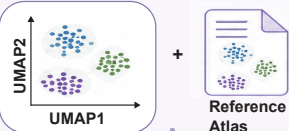


Automated Data Processing

Alignment → QC → Doublet → Filtering → Normalization → Clustering

Hierarchical Cell Annotation

Reference Annotation



SingleR

Major Cell lineages



Marker Refinement

Marker expression

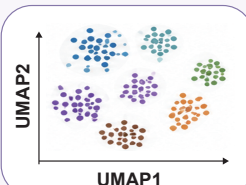


Rule-based refinement
using curated markers

Subtype refinement



Final Cell Identities

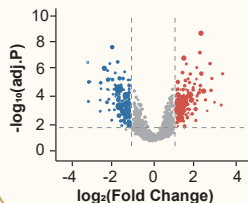


Subtype refinement

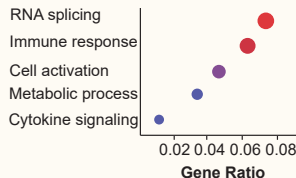


Integrated Downstream Analyses

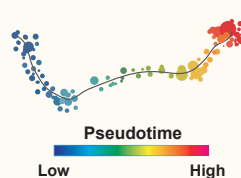
Differential Expression



Functional Enrichment



Trajectory Inference



Cell-Cell Communication

